SG No. 139 D_{4h}^{17} $I4/mmm$ [tetragonal]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$

Table 1: Wyckoff site: `2a`, site symmetry: `4/mmm`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/mmm`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `mmm.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,5,6,9,10,13,14]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,4,7,8,11,12,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4m2`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[3,4,5,6,9,10,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `4mm`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,13,14,15,16]`
2	$[0,\ 0,\ -z]$	`[5,6,7,8,9,10,11,12]`

Table 6: Wyckoff site: `8f`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,8,9,16]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,7,10,15]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,6,11,14]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[4,5,12,13]`

Table 7: Wyckoff site: `8g`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,13,14]`

continued ...

Table 7

No.	position	mapping
2	$\left[\frac{1}{2}, 0, z\right]$	[3,4,15,16]
3	$\left[0, \frac{1}{2}, -z\right]$	[5,6,9,10]
4	$\left[\frac{1}{2}, 0, -z\right]$	[7,8,11,12]

Table 8: Wyckoff site: 8h, site symmetry: m.2m

No.	position	mapping
1	$[x, x, 0]$	[1,7,10,16]
2	$[-x, -x, 0]$	[2,8,9,15]
3	$[-x, x, 0]$	[3,5,12,14]
4	$[x, -x, 0]$	[4,6,11,13]

Table 9: Wyckoff site: 8i, site symmetry: m2m.

No.	position	mapping
1	$[x, 0, 0]$	[1,6,10,13]
2	$[-x, 0, 0]$	[2,5,9,14]
3	$[0, x, 0]$	[3,7,12,16]
4	$[0, -x, 0]$	[4,8,11,15]

Table 10: Wyckoff site: 8j, site symmetry: m2m.

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,6,10,13]
2	$\left[-x, \frac{1}{2}, 0\right]$	[2,5,9,14]
3	$\left[\frac{1}{2}, x, 0\right]$	[3,7,12,16]
4	$\left[\frac{1}{2}, -x, 0\right]$	[4,8,11,15]

Table 11: Wyckoff site: 16k, site symmetry: ..2

No.	position	mapping
1	$\left[x, x+\frac{1}{2}, \frac{1}{4}\right]$	[1,7]
2	$\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$	[2,8]
3	$\left[\frac{1}{2}-x, x, \frac{1}{4}\right]$	[3,5]
4	$\left[x+\frac{1}{2}, -x, \frac{1}{4}\right]$	[4,6]
5	$\left[-x, \frac{1}{2}-x, \frac{3}{4}\right]$	[9,15]
6	$\left[x, x+\frac{1}{2}, \frac{3}{4}\right]$	[10,16]
7	$\left[x+\frac{1}{2}, -x, \frac{3}{4}\right]$	[11,13]
8	$\left[\frac{1}{2}-x, x, \frac{3}{4}\right]$	[12,14]

Table 12: Wyckoff site: `16l`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,10]`
2	$[-x,\ -y,\ 0]$	`[2,9]`
3	$[-y,\ x,\ 0]$	`[3,12]`
4	$[y,\ -x,\ 0]$	`[4,11]`
5	$[-x,\ y,\ 0]$	`[5,14]`
6	$[x,\ -y,\ 0]$	`[6,13]`
7	$[y,\ x,\ 0]$	`[7,16]`
8	$[-y,\ -x,\ 0]$	`[8,15]`

Table 13: Wyckoff site: `16m`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,16]`
2	$[-x,\ -x,\ z]$	`[2,15]`
3	$[-x,\ x,\ z]$	`[3,14]`
4	$[x,\ -x,\ z]$	`[4,13]`
5	$[-x,\ x,\ -z]$	`[5,12]`
6	$[x,\ -x,\ -z]$	`[6,11]`
7	$[x,\ x,\ -z]$	`[7,10]`
8	$[-x,\ -x,\ -z]$	`[8,9]`

Table 14: Wyckoff site: `16n`, site symmetry: `.m.`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,14]`
2	$[0,\ -y,\ z]$	`[2,13]`
3	$[-y,\ 0,\ z]$	`[3,15]`
4	$[y,\ 0,\ z]$	`[4,16]`
5	$[0,\ y,\ -z]$	`[5,10]`
6	$[0,\ -y,\ -z]$	`[6,9]`
7	$[y,\ 0,\ -z]$	`[7,11]`
8	$[-y,\ 0,\ -z]$	`[8,12]`

Table 15: Wyckoff site: `32o`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$[-y,\ x,\ z]$	`[3]`
4	$[y,\ -x,\ z]$	`[4]`
5	$[-x,\ y,\ -z]$	`[5]`
6	$[x,\ -y,\ -z]$	`[6]`
7	$[y,\ x,\ -z]$	`[7]`

continued ...

Table 15

No.	position	mapping
8	$[-y, -x, -z]$	`[8]`
9	$[-x, -y, -z]$	`[9]`
10	$[x, y, -z]$	`[10]`
11	$[y, -x, -z]$	`[11]`
12	$[-y, x, -z]$	`[12]`
13	$[x, -y, z]$	`[13]`
14	$[-x, y, z]$	`[14]`
15	$[-y, -x, z]$	`[15]`
16	$[y, x, z]$	`[16]`